Predicting SpaceX Falcon 9 First-Stage Landings

IBM Applied Data Science Capstone

Dimash · SDU University, Almaty · October 2026

https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX



Executive Summary

Methods

- SpaceX REST API and Wikipedia web scraping
- Wrangling into a binary landing label (Class)
- EDA with charts and SQL
- Folium map and Plotly Dash dashboard
- Four tuned classifiers

Key results

- Success rose from 0% (2013) to 84% (2020)
- Orbit, launch site and payload all shift the odds
- SVM, KNN and Logistic Regression tie at 83.3% test accuracy

Bottom line

- Landing success can be predicted from pre-launch features
- A predicted landing is a predicted reuse, so a cost estimate follows
- Competing bidders can price against SpaceX with it

90 launches analysed

83.3% test accuracy of the best models

66.7% overall landing success rate

Introduction

Project background

- SpaceX advertises Falcon 9 launches at about $62M; other providers charge $165M or more
- Most of the saving comes from reusing the first stage
- If we can predict whether stage 1 lands, we can estimate the cost of a launch
- That is valuable for any company that wants to bid against SpaceX

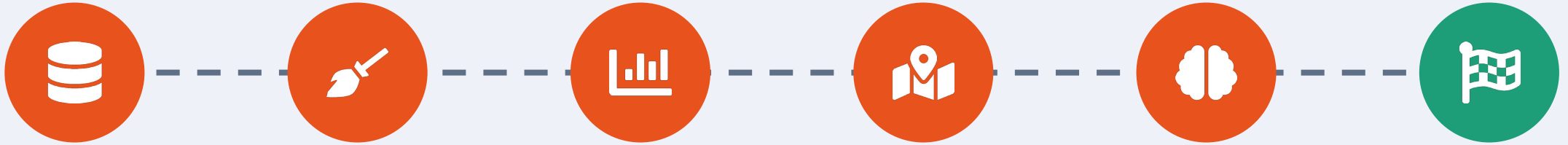
PROBLEM STATEMENT

Can we predict whether the Falcon 9 first stage will land successfully, using only information known before launch?

QUESTIONS WE ANSWER

- Which features drive landing success?
- Does success improve over time and by site?
- Which classifier predicts best?

Project Journey



Collect

SpaceX API and Wikipedia scraping

Wrangle

Clean data, create the Class label

Explore

Charts and SQL queries

Visualize

Folium map and Dash dashboard

Predict

Four classifiers, tuned

Conclude

Findings and insights

Each step feeds the next: one question, one dataset, one answer.

Methodology

Data collection · Wrangling · EDA · Interactive analytics · Modelling

Data Collection: SpaceX API

1

Request launches

GET /v4/launches/past with the requests library

2

Parse JSON

response.json() then json_normalize() into a DataFrame

3

Resolve IDs

Helper functions call /rockets, /launchpads, /payloads and /cores

4

Falcon 9 only

Drop Falcon 1, multi-core and multi-payload launches

5

Save dataset

Flight number, site, orbit, payload, outcome per launch

KEY PHRASES

requests.get() | json_normalize | Falcon 9 filter | to_csv

GitHub https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection: Web Scraping

1

Request page

Download the Wikipedia "List of Falcon 9 and Falcon Heavy launches" snapshot

→

2

Parse HTML

BeautifulSoup builds the document tree

→

3

Find tables

Select the launch tables by their wikitable class

→

4

Extract columns

Flight no., date, booster, site, payload, orbit, outcome

→

5

Build DataFrame

Combine all rows and save to CSV

KEY PHRASES

requests.get() BeautifulSoup find_all("table") pd.DataFrame



GitHub https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

1

Load data

Read the collected launch table

2

Missing values

PayloadMass filled with the mean; LandingPad nulls kept

3

Count

Launches per site, per orbit, and mission outcomes

4

Create Class

Turn the landing outcome into a 0 / 1 label

Landing outcome → Class

Class 1

- True Ocean
- True RTLS
- True ASDS

Class 0

- False Ocean
- False RTLS
- False ASDS
- None None

Why a binary label?

Classification needs one target: 1 means the first stage landed, 0 means it did not.



GitHub https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb

EDA with Data Visualization: Methodology

Chart	Variables	Purpose
Scatter	Flight Number × Launch Site	Does experience at a site improve landing success?
Scatter	Payload Mass × Launch Site	Do some sites handle heavier payloads?
Bar	Success rate × Orbit	Which orbits are easiest to land from?
Scatter	Flight Number × Orbit	Does the learning effect hold in each orbit?
Scatter	Payload Mass × Orbit	How do payload and orbit interact?
Line	Success rate × Year	Is the overall trend improving?



Feature engineering: one-hot encode Orbit, LaunchSite, LandingPad and Serial so the models can use them.



GitHub https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/SpaceX_Capstone_Analysis.ipynb

EDA with SQL: Methodology

1. Display the unique launch site names
2. Show 5 records for sites starting with "CCA"
3. Total payload mass carried for NASA (CRS)
4. Average payload mass of booster F9 v1.1
5. Date of the first successful ground-pad landing
6. Drone-ship boosters with payload 4,000 to 6,000 kg
7. Count successful vs failed mission outcomes
8. Boosters that carried the maximum payload
9. 2015 drone-ship failures: month, booster, site
10. Rank landing outcomes, 2010-06-04 to 2017-03-20



GitHub https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/SpaceX_Capstone_Analysis.ipynb

Interactive Visual Analytics: Methodology

Folium map

- A marker and circle for every launch site
- MarkerCluster: green = success, red = failure
- MousePosition to read coordinates
- PolyLine and distance label to coast, railway, highway and city

GitHub
https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/lab_jupyter_launch_site_location_completed.ipynb

Plotly Dash app

- Dropdown: all sites or one site
- Pie chart: successful launches by site
- Range slider for payload mass
- Scatter: payload vs outcome by booster version
- Callbacks link controls to both charts

GitHub
https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/spacex_dash_app.py

Predictive Analysis: Methodology

1

Standardize

StandardScaler on the features X; target is Class

2

Split

train_test_split: 80% train, 20% test

3

Tune

GridSearchCV with 10-fold cross-validation

4

Evaluate

Test accuracy and confusion matrix

5

Select

Highest accuracy is the best model

Logistic Regression

TUNED

C, penalty, solver

SVM

TUNED

kernel, C, gamma

Decision Tree

TUNED

criterion, max_depth, splitter

KNN

TUNED

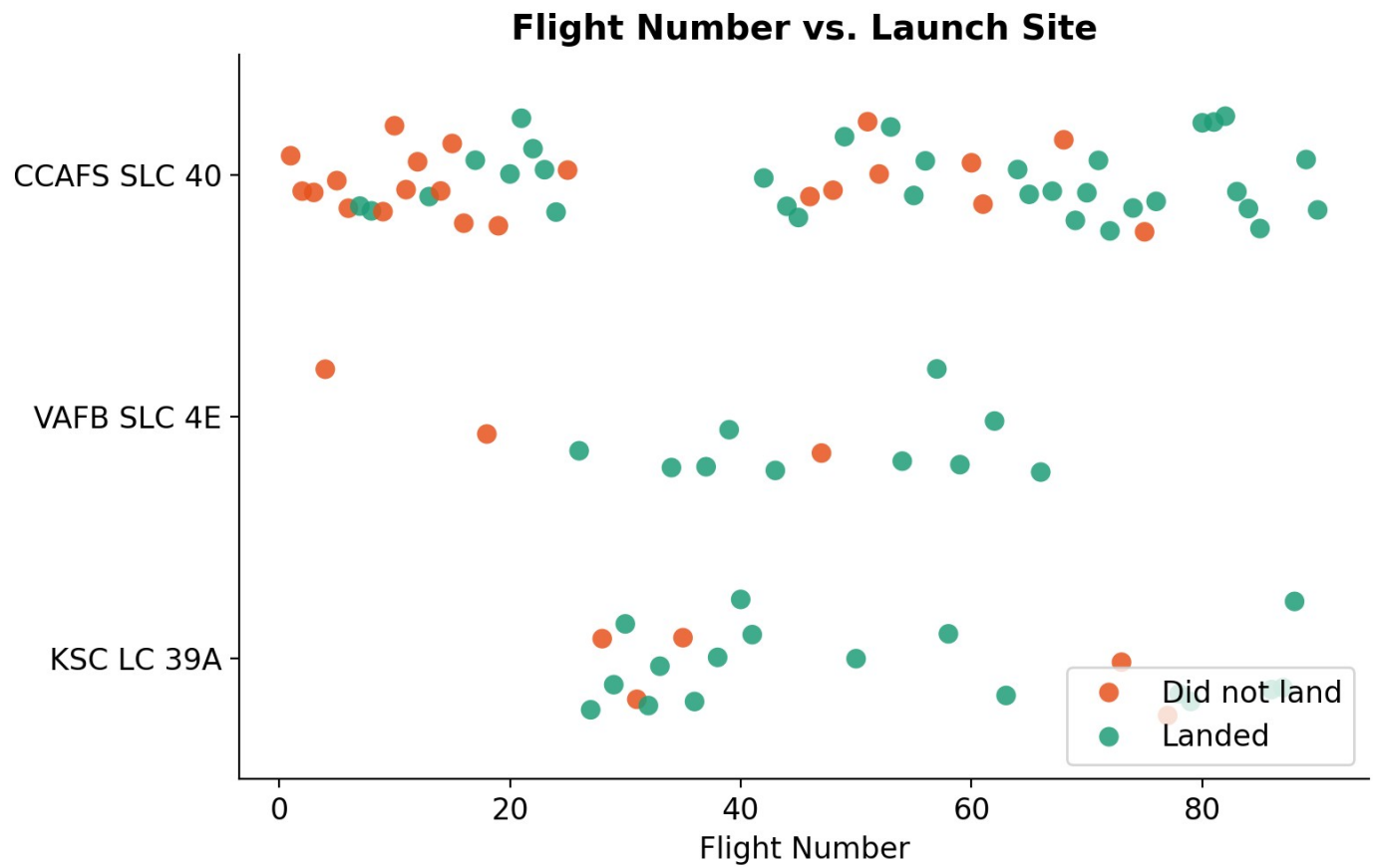
n_neighbors, p, algorithm

GitHub https://github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/SpaceX_Capstone_Analysis.ipynb

Results

EDA · SQL · Folium · Plotly Dash · Classification

EDA: Flight Number vs. Launch Site



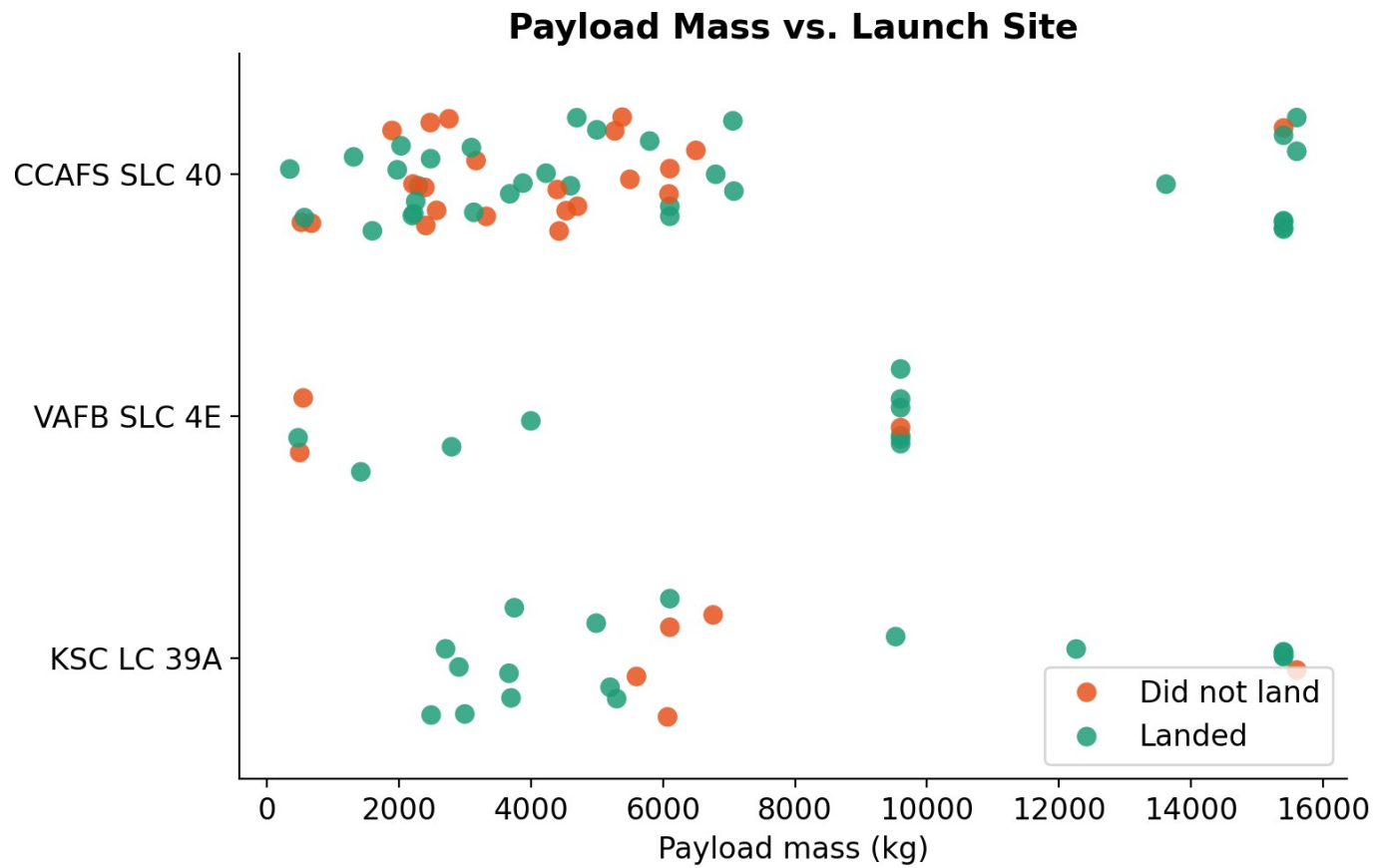
WHAT IT SHOWS

Each dot is one launch, coloured by landing outcome and grouped by launch site.

INSIGHT

Success rises in the second half of the launch history at every site (flights 1 to 45 vs 46 to 90): CCAFS SLC 40 from 46% to 72%, VAFB SLC 4E from 71% to 83%, KSC LC 39A from 75% to 80%.

EDA: Payload vs. Launch Site



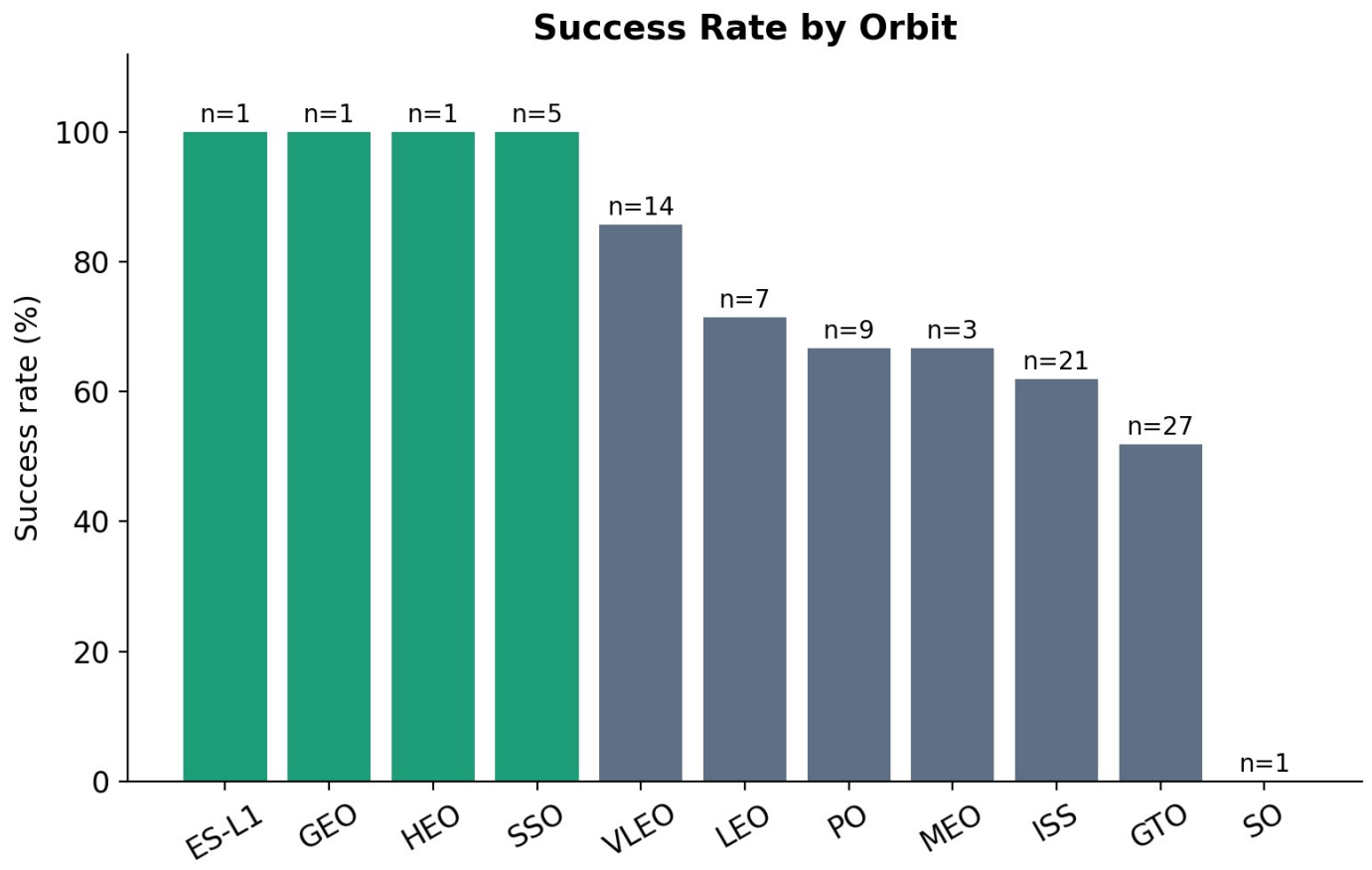
WHAT IT SHOWS

Payload mass (kg) against launch site, coloured by outcome.

INSIGHT

The heaviest payloads (up to 15,600 kg) fly from CCAFS SLC 40 and KSC LC 39A. VAFB SLC 4E has no launch above 10,000 kg (maximum 9,600 kg).

EDA: Success Rate by Orbit



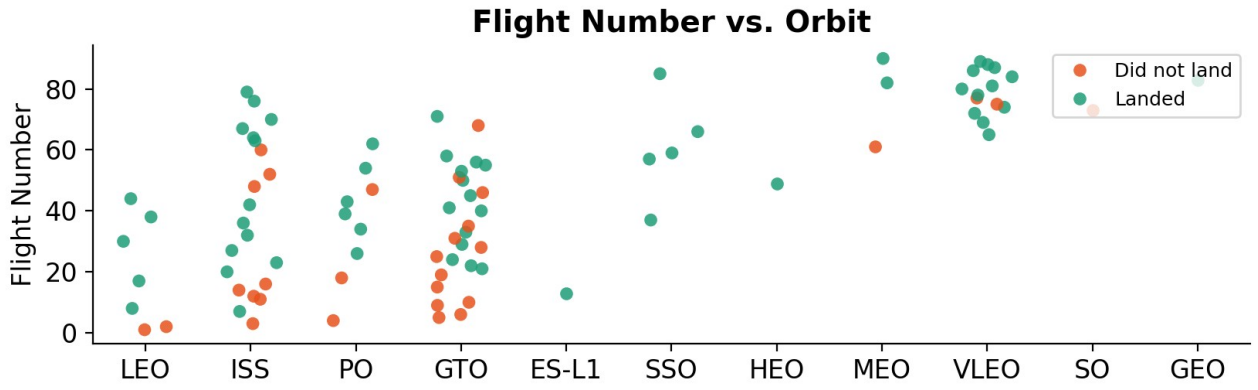
WHAT IT SHOWS

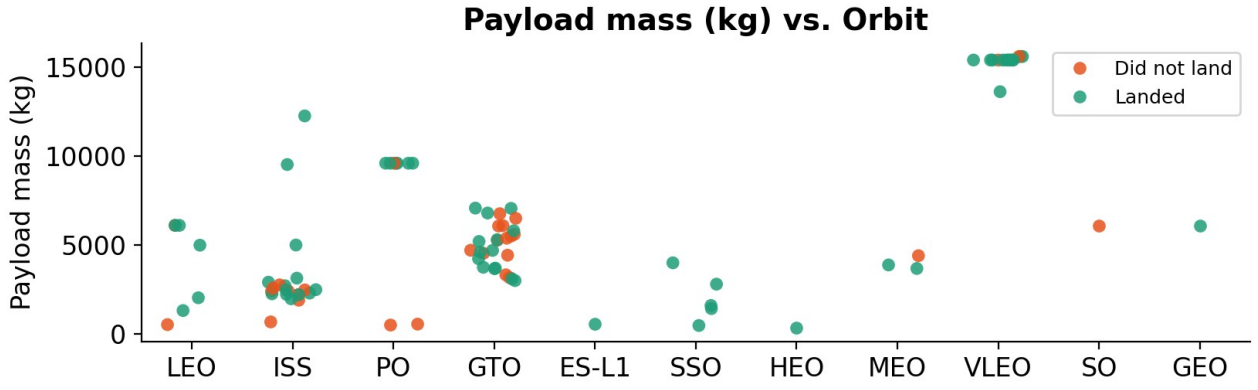
The share of launches per orbit whose first stage landed, with the number of launches (n).

INSIGHT

ES-L1, GEO, HEO and SSO reach 100%, but three of them have a single launch. Among well-sampled orbits VLEO leads (86%, n=14) and GTO is hardest (52%, n=27).

EDA: Flight Number and Payload vs. Orbit





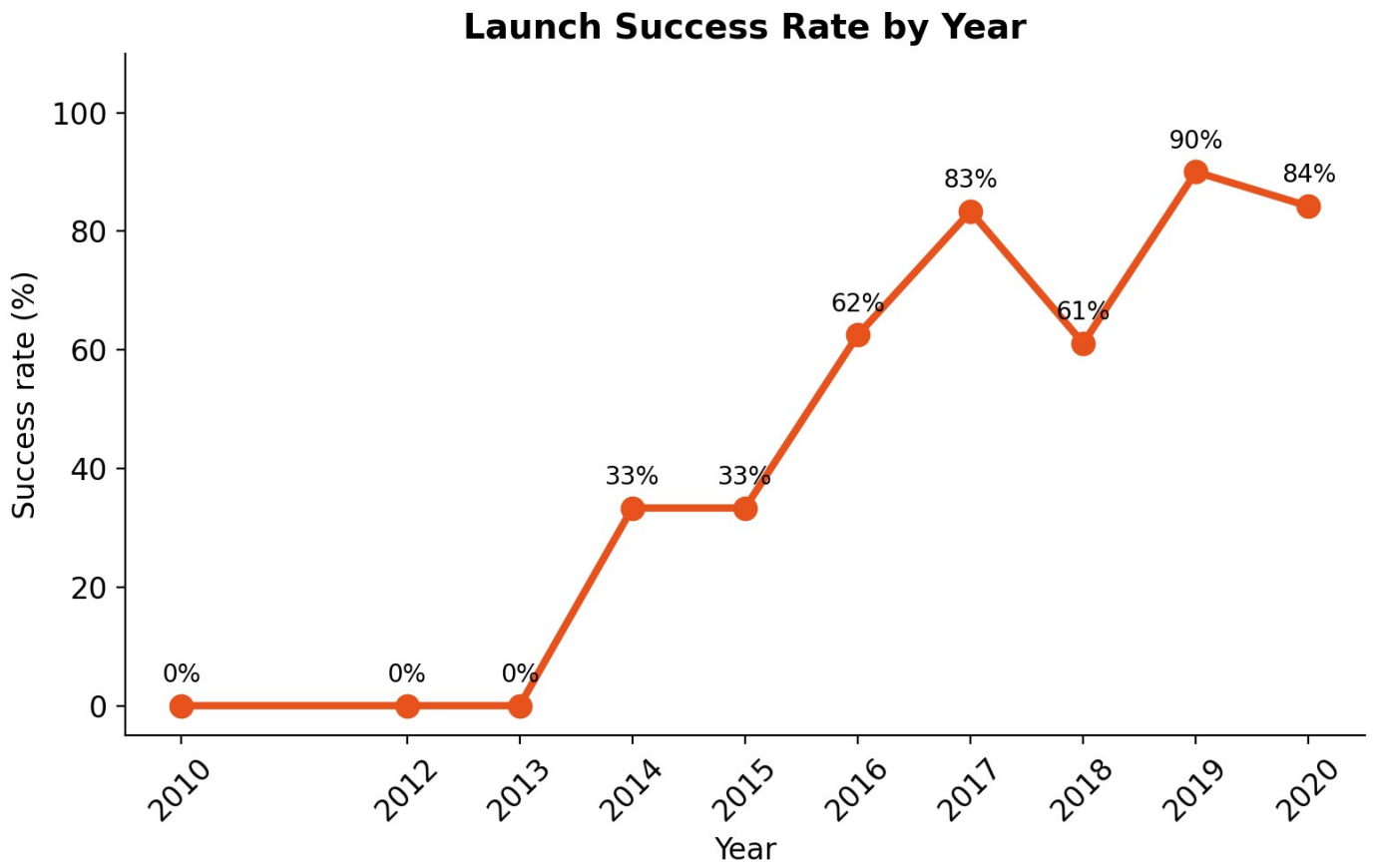
WHAT IT SHOWS

Two scatter plots comparing orbits by experience (flight number) and by payload mass.

INSIGHT

In LEO, success grows with flight number (correlation 0.72); in GTO the link is weaker (0.43). The 15 launches above 10,000 kg (14 in VLEO, 1 in ISS) landed 87% of the time.

EDA: Yearly Launch Success Trend



WHAT IT SHOWS

The share of launches that landed in each year.

INSIGHT

Success climbs from 0% in 2013 to 84% in 2020, peaking at 90% in 2019, with a dip to 61% in 2018.

EDA with SQL: Launch Sites

Q1 Display the unique launch site names

SELECT DISTINCT Launch_Site FROM SPACEXTBL

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Q2 5 records where launch sites begin with "CCA"

SELECT Date, Booster_Version, Launch_Site, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA %' LIMIT 5

Date	Booster_Version	Launch_Site	Payload (kg)
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	0
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	0
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	525
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	500
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	677

WHAT IT SHOWS

The distinct launch sites, and a preview of launches from sites starting with "CCA".

INSIGHT

Four distinct sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A and CCAFS SLC-40. The first two launches (2010) carried 0 kg of payload.

EDA with SQL: Payload and First Landing

Q3 Total payload mass carried by boosters launched by NASA (CRS)

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_kg FROM
SPACEXTBL WHERE Customer = 'NASA (CRS)'
```

45,596 kg

Q4 Average payload mass carried by booster version F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload_kg FROM SPACEXTBL
WHERE Booster_Version = 'F9 v1.1'
```

2,928.4 kg

Q5 Date of the first successful landing on a ground pad

```
SELECT MIN(Date) AS first_ground_pad_landing FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)'
```

2015-12-22

WHAT IT SHOWS

Total payload for NASA (CRS), the average payload of F9 v1.1, and the date of the first ground-pad landing.

INSIGHT

NASA (CRS) missions carried 45,596 kg in total. F9 v1.1 boosters averaged 2,928.4 kg. The first successful ground-pad landing was on 22 December 2015.

EDA with SQL: Boosters and Mission Outcomes

Q6 Boosters with drone-ship success and payload between 4,000 and 6,000 kg

```
SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Q7 Total number of successful and failed mission outcomes

```
SELECT Mission_Outcome, COUNT(*) AS total FROM SPACEXTBL GROUP BY Mission_Outcome
```

Mission_Outcome	Total
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

WHAT IT SHOWS

Boosters that landed on a drone ship with a mid-range payload, and the count of each mission outcome.

INSIGHT

Four Falcon 9 FT boosters landed on a drone ship with payloads of 4,000 to 6,000 kg. Of 101 missions, 99 succeeded, 1 had an unclear payload status and 1 failed in flight.

EDA with SQL: Heaviest Payload and 2015 Failures

Q8 Boosters that carried the maximum payload mass

SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)

Booster_Version	Booster_Version
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

Q9 2015 failed drone-ship landings: month, booster, launch site

SELECT substr(Date, 6, 2) AS month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date, 1, 4) = '2015'

month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

WHAT IT SHOWS

Boosters that carried the maximum payload (15,600 kg), and the failed drone-ship landings of 2015.

INSIGHT

12 booster flights carried the maximum payload, all Block 5 (B5). Two drone-ship landings failed in 2015: January (B1012) and April (B1015), both from CCAFS LC-40.

EDA with SQL: Landing Outcome Ranking

Q10 Rank landing outcomes between 2010-06-04 and 2017-03-20 (descending)

```
SELECT Landing_Outcome, COUNT(*) AS total FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY total DESC
```

Landing_Outcome	Total
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

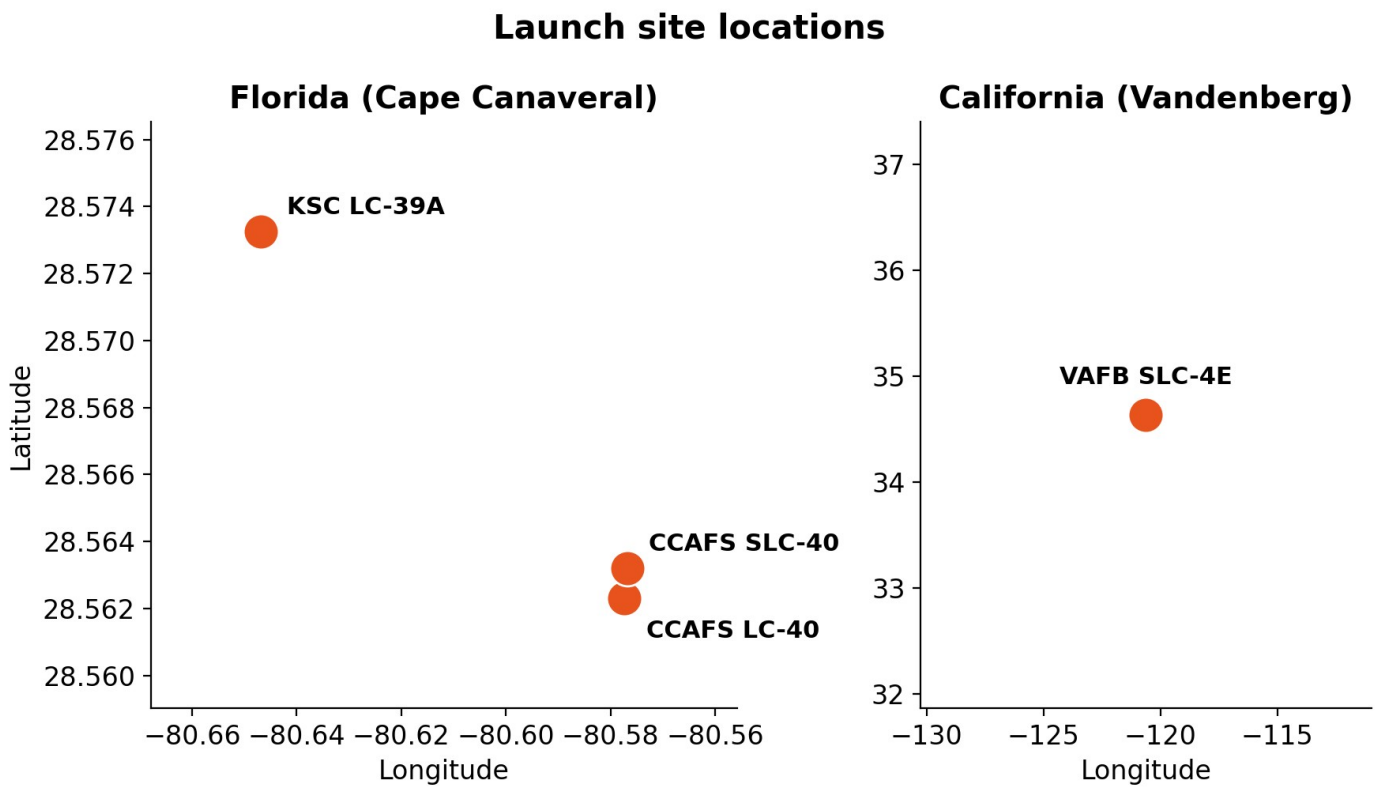
WHAT IT SHOWS

Landing outcomes between 2010-06-04 and 2017-03-20, counted and ranked in descending order.

INSIGHT

"No attempt" was the most common outcome (10). Drone-ship successes and failures tied at 5 each, ahead of ground-pad successes (3).

Folium Map: Launch Site Locations



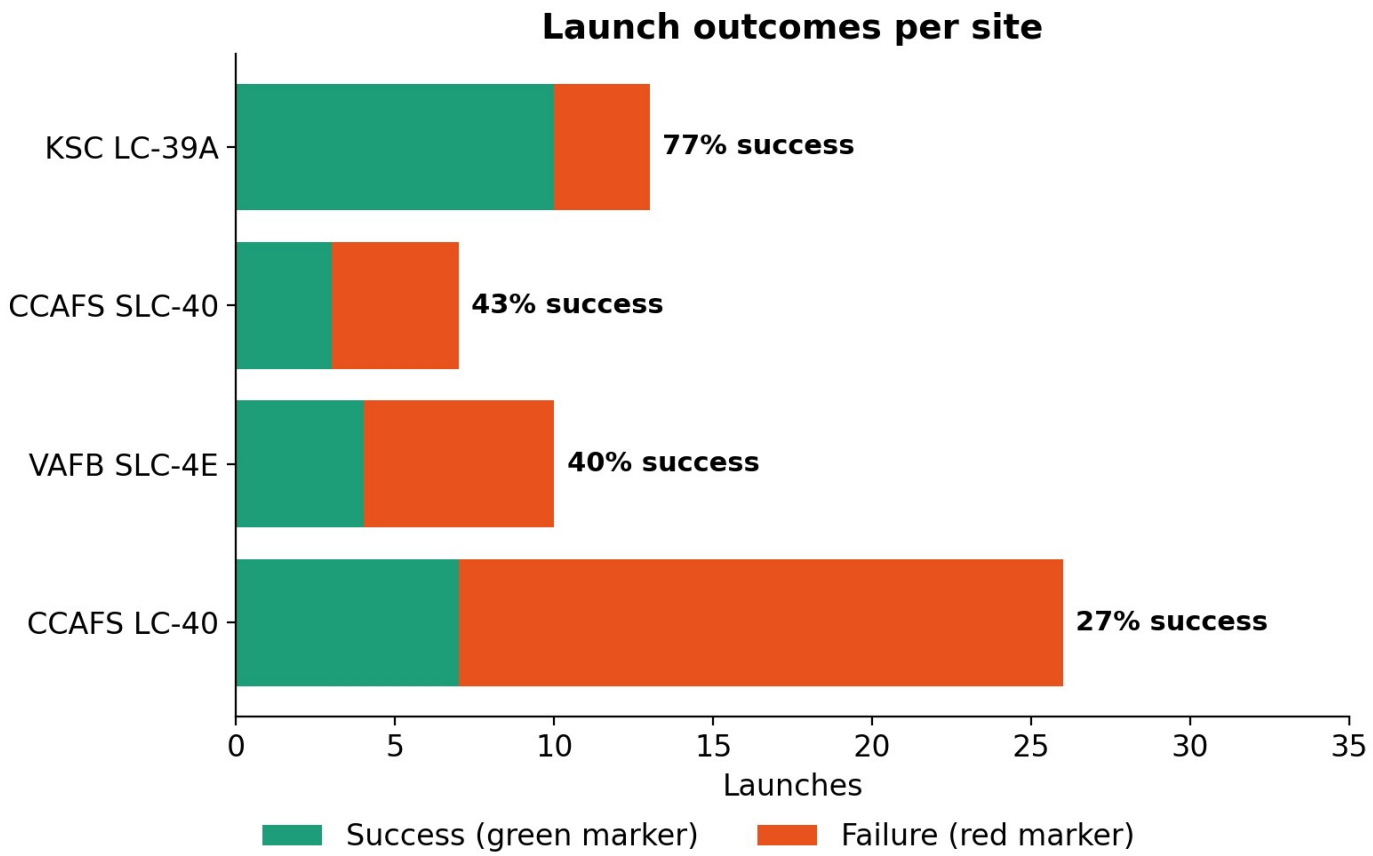
WHAT IT SHOWS

Launch sites plotted by latitude and longitude, as marked on the Folium map. Static view; the interactive map is in the notebook.

INSIGHT

Three sites sit together at Cape Canaveral, Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A). The fourth, VAFB SLC-4E, is in California. All four are on the US coast.

Folium Map: Launch Outcomes per Site



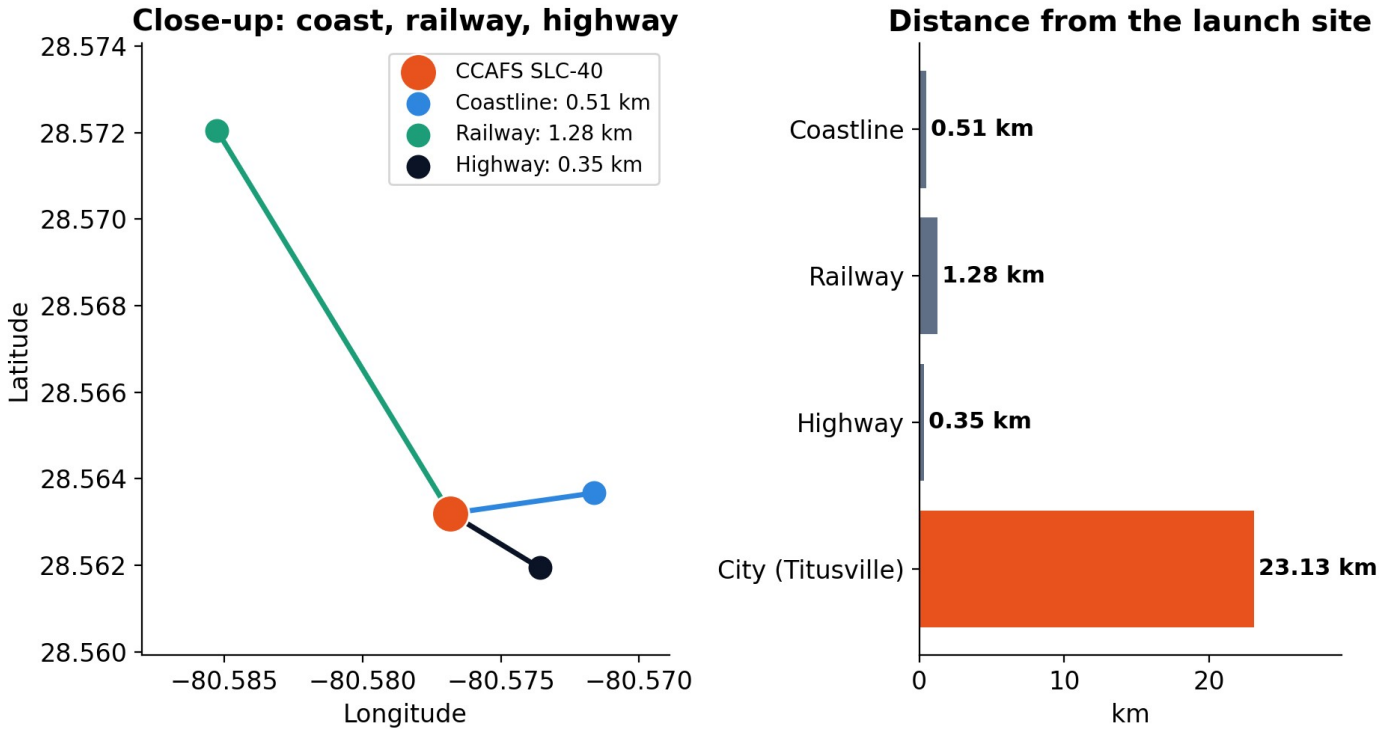
WHAT IT SHOWS

Green markers are successful landings and red markers are failures at each site. Static view of the marker counts.

INSIGHT

KSC LC-39A has the best record: 10 of 13 launches landed (77%). CCAFS LC-40 has the weakest, 7 of 26 (27%), and hosts the earliest launches.

Folium Map: Proximity Analysis



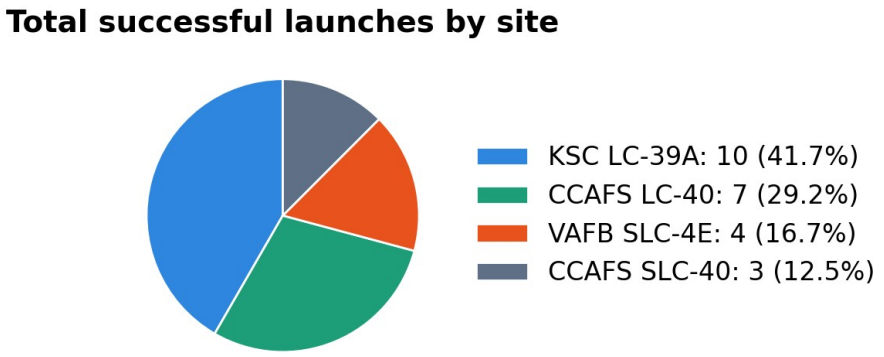
WHAT IT SHOWS

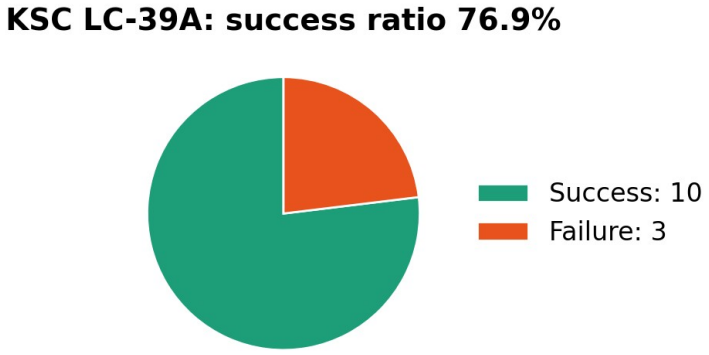
Distances from CCAFS SLC-40 to the nearest coastline, railway, highway and city, as drawn with lines on the Folium map.

INSIGHT

The site is 0.51 km from the coast, 1.28 km from a railway and 0.35 km from a highway, but 23 km from the nearest city (Titusville). Proximity points are approximate.

Plotly Dash: Launch Success by Site





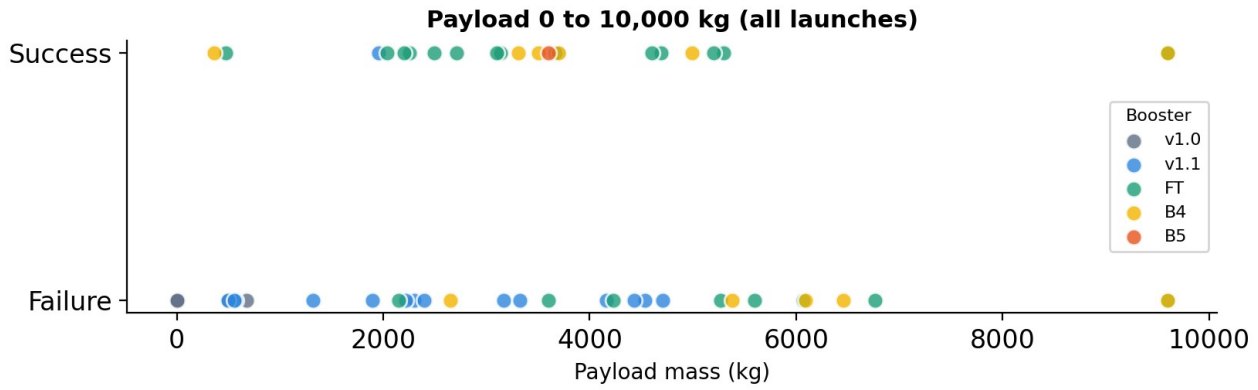
WHAT IT SHOWS

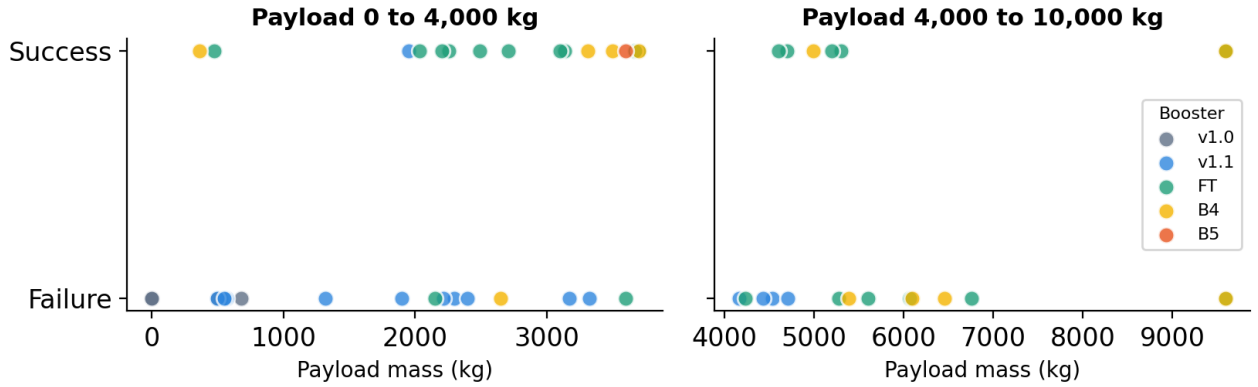
The two pie charts of the dashboard: successful launches per site (top), and success vs failure for the site with the best ratio (bottom).

INSIGHT

KSC LC-39A holds the largest share of successful launches (10 of 24, 41.7%) and the best success ratio (76.9%).

Plotly Dash: Payload vs. Outcome





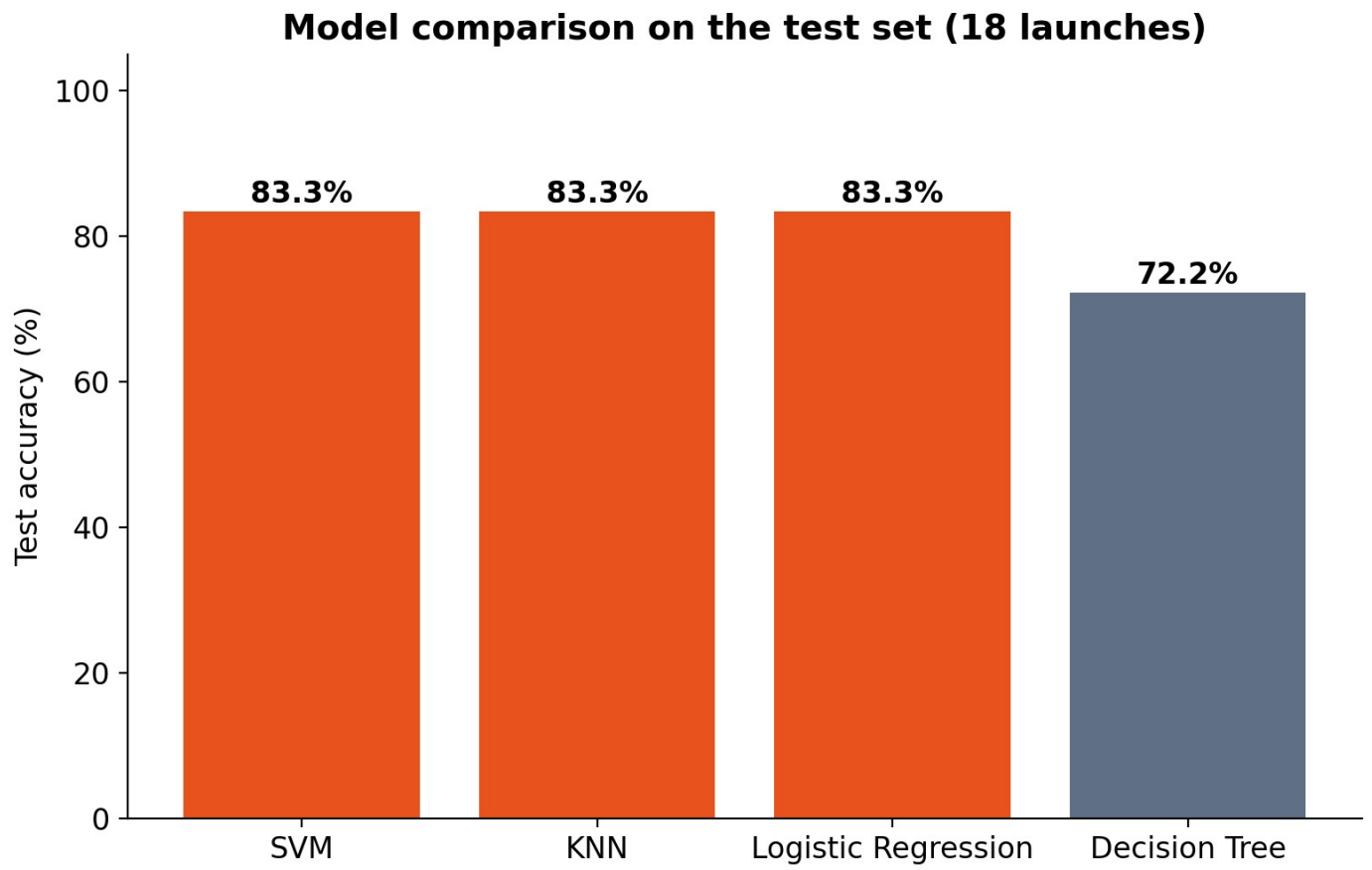
WHAT IT SHOWS

Payload mass against outcome, coloured by booster version, filtered with the payload range slider.

INSIGHT

Launches with 0 to 4,000 kg landed 47% of the time (n=34) vs 36% for 4,000 to 10,000 kg (n=22). By booster, FT lands 67% (n=24); v1.0 and v1.1 rarely landed (0% and 7%).

Predictive Analysis: Model Comparison



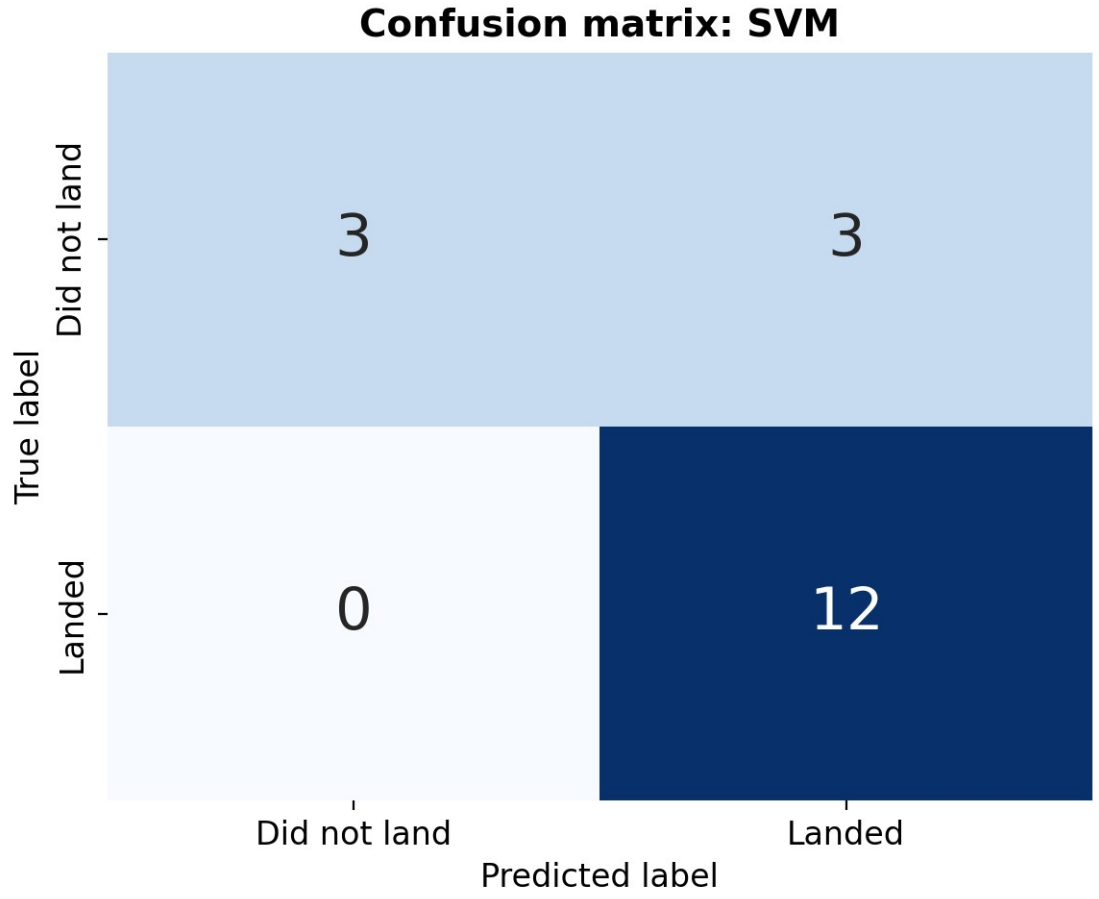
WHAT IT SHOWS

Test-set accuracy of each tuned classifier: Logistic Regression, SVM, Decision Tree and KNN.

INSIGHT

SVM, KNN and Logistic Regression tie at 83.3% test accuracy; the Decision Tree reaches 72.2%. SVM is selected for its best cross-validation accuracy (84.8%).

Predictive Analysis: Confusion Matrix



WHAT IT SHOWS

True vs predicted landing outcomes for the best model on the 18 test launches.

INSIGHT

12 true positives, 3 true negatives, 3 false positives and 0 false negatives. The model never misses a landing but sometimes predicts a landing that fails.

Innovative Insights

Reuse pays for itself

FINDING

A landed first stage supports the gap between $165M and $62M launch prices.

SO WHAT

Expected saving ≈ P(landing) × ~$100M. At the historical 66.7% landing rate that is about $67M per launch; a model probability makes it launch-specific.

Experience compounds

FINDING

Success rises from 0% (2013) to 84% (2020); in LEO it tracks flight number (correlation 0.72).

SO WHAT

Early flights drag down naive averages; model booster maturity, not just orbit.

Context beats one factor

FINDING

GTO lands 52% of the time vs 86% for VLEO, and KSC LC-39A lands 77% vs 27% at CCAFS LC-40.

SO WHAT

No single factor explains a landing; the models combine orbit, site, payload and booster.

Conclusion

Landing success is predictable

SVM, KNN and Logistic Regression reach 83.3% test accuracy.

Success improves with experience

The yearly landing rate rose from 0% in 2013 to 84% in 2020.

Context matters

Orbit, launch site, payload and booster version all shift the odds.

Limits and next steps

Only 90 launches and 18 test samples. Add more data, weather and reuse-count features, and probability outputs.

Appendix: GitHub Links

Deliverable	Link
Repository (all notebooks and code)	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX
Data collection: SpaceX API	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/jupyter-labs-spacex-data-collection-api.ipynb
Data collection: web scraping	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/jupyter-labs-webscraping.ipynb
Data wrangling	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb
EDA with SQL	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/SpaceX_Capstone_Analysis.ipynb
EDA with data visualization	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/SpaceX_Capstone_Analysis.ipynb
Folium map	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/lab_jupyter_launch_site_location_completed.ipynb
Plotly Dash app	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/spacex_dash_app.py
Predictive analysis	github.com/itsdimash/IBM-Data-Science-Capstone-SpaceX/blob/main/SpaceX_Capstone_Analysis.ipynb

Appendix: Model Details

Model	Best hyperparameters (GridSearchCV, cv=10)	CV accuracy	Test accuracy
Logistic Regression	C=1, penalty=l2, solver=lbfgs	82.1%	83.3%
SVM	C=1, gamma=0.0316, kernel=sigmoid	84.8%	83.3%
Decision Tree	criterion=gini, max_depth=10, max_features=sqrt, min_samples_leaf=1, min_samples_split=2, splitter=random	83.4%	72.2%
KNN	algorithm=auto, n_neighbors=6, p=1	83.4%	83.3%

Split: 72 training and 18 test launches (test_size 0.2, random_state 2), features standardized with StandardScaler. With 18 test launches, one launch changes accuracy by 5.6 points, so the tie between three models should be read with care.